



Divisibility Proof by Induction

Practice Question Set

Proving Divisibility Statements using the Principle of
Mathematical Induction

9 Questions • Fully Worked Answers

Instructions: Attempt each proof yourself first. The full worked answer is given directly beneath each question, boxed in light blue, so you can check your working immediately.

Question 3(a)

Use the principle of mathematical induction to prove that for all $n \in \mathbb{Z}^+$:

$$n^3 + 2n \text{ is divisible by } 3.$$

Answer

Let P_n be the proposition: $n^3 + 2n$ is divisible by 3, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $1^3 + 2(1) = 3$, which is divisible by 3. $\therefore P_1$ is true.

(2) If P_k is true, then $k^3 + 2k = 3A$ where $A \in \mathbb{Z}$.

Now

$$\begin{aligned} & (k+1)^3 + 2(k+1) \\ &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\ &= (k^3 + 2k) + 3k^2 + 3k + 3 \\ &= 3A + 3k^2 + 3k + 3 \quad \{\text{using } P_k\} \\ &= 3(A + k^2 + k + 1) \\ &\text{where } (A + k^2 + k + 1) \in \mathbb{Z} \\ &\text{as } A \in \mathbb{Z} \text{ and } k \in \mathbb{Z}^+. \end{aligned}$$

Thus $(k+1)^3 + 2(k+1)$ is divisible by 3 if $k^3 + 2k$ is divisible by 3.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 3(b)

Use the principle of mathematical induction to prove that for all $n \in \mathbb{Z}^+$:

$$7^n - 1 \text{ is divisible by } 3.$$

Answer

Let P_n be the proposition: $7^n - 1$ is divisible by 3, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $7^1 - 1 = 6$, which is divisible by 3. $\therefore P_1$ is true.

(2) If P_k is true, then $7^k - 1 = 3A$ for some $A \in \mathbb{Z}^+$.

Now

$$\begin{aligned} 7^{k+1} - 1 &= 7 \times 7^k - 1 \\ &= 7(3A + 1) - 1 \quad \{\text{using } P_k\} \\ &= 21A + 6 \\ &= 3(7A + 2) \\ &\quad \text{where } (7A + 2) \in \mathbb{Z} \\ &\quad \text{as } A \in \mathbb{Z}^+. \end{aligned}$$

Thus $7^{k+1} - 1$ is divisible by 3 if $7^k - 1$ is divisible by 3.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 3(c)

Use the principle of mathematical induction to prove that for all $n \in \mathbb{Z}^+$:

$$8^n - 3^n \text{ is divisible by } 5.$$

Answer

Let P_n be the proposition: $8^n - 3^n$ is divisible by 5, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $8^1 - 3^1 = 5$, which is divisible by 5. $\therefore P_1$ is true.

(2) If P_k is true, then $8^k - 3^k = 5A$ for some $A \in \mathbb{Z}^+$.

Now

$$\begin{aligned} 8^{k+1} - 3^{k+1} &= 8(8^k - 3^k) + 8 \times 3^k - 3 \times 3^k \\ &= 8(5A) + 5 \times 3^k \quad \{\text{using } P_k\} \\ &= 5(8A + 3^k) \end{aligned}$$

$$\text{where } (8A + 3^k) \in \mathbb{Z}$$

$$\text{as } A \in \mathbb{Z}^+ \text{ and } k \in \mathbb{Z}^+.$$

Thus $8^{k+1} - 3^{k+1}$ is divisible by 5 if $8^k - 3^k$ is divisible by 5.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 3(d)

Use the principle of mathematical induction to prove that for all $n \in \mathbb{Z}^+$:

$$7^n - 4^n - 3^n \text{ is divisible by } 12.$$

Answer

Let P_n be the proposition: $7^n - 4^n - 3^n$ is divisible by 12, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $7^1 - 4^1 - 3^1 = 0$, which is divisible by 12. $\therefore P_1$ is true.

(2) If P_k is true, then $7^k - 4^k - 3^k = 12A$ where $A \in \mathbb{Z}$.

Now

$$\begin{aligned} &7^{k+1} - 4^{k+1} - 3^{k+1} \\ &= 7(7^k) - 4(4^k) - 3(3^k) \\ &= 7[12A + 4^k + 3^k] - 4(4^k) - 3(3^k) \quad \{\text{using } P_k\} \\ &= 84A + 7(4^k) + 7(3^k) - 4(4^k) - 3(3^k) \\ &= 84A + 3(4^k) + 4(3^k) \\ &= 84A + 3 \times 4 \times 4^{k-1} + 4 \times 3 \times 3^{k-1} \\ &= 12(7A + 4^{k-1} + 3^{k-1}) \\ &\text{where } (7A + 4^{k-1} + 3^{k-1}) \in \mathbb{Z} \\ &\text{as } A \in \mathbb{Z} \text{ and } k \in \mathbb{Z}^+. \end{aligned}$$

Thus $7^{k+1} - 4^{k+1} - 3^{k+1}$ is divisible by 12 if $7^k - 4^k - 3^k$ is divisible by 12.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 3(e)

Use the principle of mathematical induction to prove that for all $n \in \mathbb{Z}^+$:

$$7^n + 2 \times 4^n \text{ is divisible by } 3.$$

Answer

Let P_n be the proposition: $7^n + 2 \times 4^n$ is divisible by 3, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $7^1 + 2 \times 4^1 = 15$, which is divisible by 3. $\therefore P_1$ is true.

(2) If P_k is true, then $7^k + 2 \times 4^k = 3A$ for some $A \in \mathbb{Z}^+$.

Now

$$\begin{aligned} 7^{k+1} + 2 \times 4^{k+1} &= 7 \times 7^k + 2 \times 4 \times 4^k \\ &= 7(3A - 2 \times 4^k) + 8 \times 4^k && \{\text{using } P_k\} \\ &= 21A - 14 \times 4^k + 8 \times 4^k \\ &= 21A - 6 \times 4^k \\ &= 3(7A - 2 \times 4^k) \\ &\text{where } (7A - 2 \times 4^k) \in \mathbb{Z}, A, k \in \mathbb{Z}^+. \end{aligned}$$

Thus $7^{k+1} + 2 \times 4^{k+1}$ is divisible by 3 if $7^k + 2 \times 4^k$ is divisible by 3.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 4(a)

Use the principle of mathematical induction to prove that for all $n \in \mathbb{Z}^+$:

$$3^{2n+4} - 2^{2n} \text{ is divisible by } 5.$$

Answer

Let P_n be the proposition: $3^{2n+4} - 2^{2n}$ is divisible by 5, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $3^{2+4} - 2^2 = 725$, which is divisible by 5. $\therefore P_1$ is true.

(2) If P_k is true, then $3^{2k+4} - 2^{2k} = 5A$ where $A \in \mathbb{Z}$.

Now

$$\begin{aligned} & 3^{2(k+1)+4} - 2^{2(k+1)} \\ &= 3^2 \times 3^{2k+4} - 2^2 \times 2^{2k} \\ &= 9(3^{2k+4} - 2^{2k}) + 5 \times 2^{2k} \\ &= 9(5A) + 5 \times 2^{2k} \quad \{\text{using } P_k\} \\ &= 5(9A + 2^{2k}) \text{ where } (9A + 2^{2k}) \in \mathbb{Z}, A \in \mathbb{Z}, k \in \mathbb{Z}^+. \end{aligned}$$

Thus $3^{2(k+1)+4} - 2^{2(k+1)}$ is divisible by 5 if $3^{2k+4} - 2^{2k}$ is divisible by 5.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 4(b)

Use the principle of mathematical induction to prove that for all $n \in \mathbb{Z}^+$:

$$3 \times 5^{2n+1} + 2^{3n+1} \text{ is divisible by } 17.$$

Answer

Let P_n be the proposition: $3 \times 5^{2n+1} + 2^{3n+1}$ is divisible by 17, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $3 \times 5^{2+1} + 2^{3+1} = 3 \times 125 + 16 = 391 = 17 \times 23$, which is divisible by 17. $\therefore P_1$ is true.

(2) If P_k is true, then $3 \times 5^{2k+1} + 2^{3k+1} = 17A$ where $A \in \mathbb{Z}$.

Now

$$\begin{aligned} & 3 \times 5^{2(k+1)+1} + 2^{3(k+1)+1} \\ &= 5^2 \times 3 \times 5^{2k+1} + 2^3 \times 2^{3k+1} \\ &= 25(3 \times 5^{2k+1} + 2^{3k+1}) - 17 \times 2^{3k+1} \\ &= 25(17A) - 17 \times 2^{3k+1} \quad \{\text{using } P_k\} \\ &= 17(25A - 2^{3k+1}) \end{aligned}$$

$$\text{where } (25A - 2^{3k+1}) \in \mathbb{Z}$$

$$\text{as } A \in \mathbb{Z} \text{ and } k \in \mathbb{Z}^+.$$

Thus $3 \times 5^{2(k+1)+1} + 2^{3(k+1)+1}$ is divisible by 17 if $3 \times 5^{2k+1} + 2^{3k+1}$ is divisible by 17.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 5

[Maximum mark: 7] Use the method of mathematical induction to prove that for all $n \in \mathbb{Z}^+$:

$$5^{2n} - 24n - 1 \text{ is divisible by } 576.$$

Answer

Let P_n be the proposition: $5^{2n} - 24n - 1$ is divisible by 576, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $5^2 - 24(1) - 1 = 25 - 24 - 1 = 0$, which is divisible by 576 (since $0 = 576 \times 0$). $\therefore P_1$ is true.

(2) If P_k is true, then $5^{2k} - 24k - 1 = 576A$ where $A \in \mathbb{Z}$, so $5^{2k} = 576A + 24k + 1$.

Now

$$\begin{aligned} & 5^{2(k+1)} - 24(k+1) - 1 \\ &= 25 \times 5^{2k} - 24k - 25 \\ &= 25(576A + 24k + 1) - 24k - 25 \quad \{\text{using } P_k\} \\ &= 14400A + 600k + 25 - 24k - 25 \\ &= 14400A + 576k \\ &= 576(25A + k) \quad \text{where } (25A + k) \in \mathbb{Z} \text{ as } A, k \in \mathbb{Z}^+. \end{aligned}$$

Thus $5^{2(k+1)} - 24(k+1) - 1$ is divisible by 576 if $5^{2k} - 24k - 1$ is divisible by 576.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}

Question 6

[Maximum mark: 8] Prove, by mathematical induction, that for all $n \in \mathbb{Z}^+$:

$$7^{8n+3} + 2 \text{ is divisible by } 5.$$

Answer

Let P_n be the proposition: $7^{8n+3} + 2$ is divisible by 5, for all $n \in \mathbb{Z}^+$.

Proof: (By the principle of mathematical induction)

(1) If $n = 1$: $7^{11} + 2 = 1\,977\,326\,745 = 5 \times 395\,465\,349$, which is divisible by 5. $\therefore P_1$ is true.

(2) If P_k is true, then $7^{8k+3} + 2 = 5A$ where $A \in \mathbb{Z}$.

Note also that $7^8 - 1 = 5\,764\,800 = 5 \times 1\,152\,960$, so $7^8 - 1 = 5C$ where $C = 1\,152\,960 \in \mathbb{Z}$.

Now

$$\begin{aligned} &7^{8(k+1)+3} + 2 \\ &= 7^{8k+3} \times 7^8 + 2 \\ &= 7^{8k+3}(7^8 - 1) + (7^{8k+3} + 2) \\ &= 7^{8k+3}(5C) + 5A \quad \{\text{using } P_k, 7^8 - 1 = 5C\} \\ &= 5(7^{8k+3}C + A) \\ &\quad \text{where } (7^{8k+3}C + A) \in \mathbb{Z} \\ &\quad \text{as } A, C \in \mathbb{Z} \text{ and } k \in \mathbb{Z}^+. \end{aligned}$$

Thus $7^{8(k+1)+3} + 2$ is divisible by 5 if $7^{8k+3} + 2$ is divisible by 5.

Since P_1 is true, and P_{k+1} is true whenever P_k is true, P_n is true for all $n \in \mathbb{Z}^+$. {principle of mathematical induction}